

EEE 4606
Project Report

Title: Self Balancing Robot using Matlab Simulink.

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Abstract

We have built a self balancing robot using matlab simulink. This robot contains 3 horizontal plates and 2 wheels. An external force is applied on the top plate. We have used a PID controller to balance the robot.

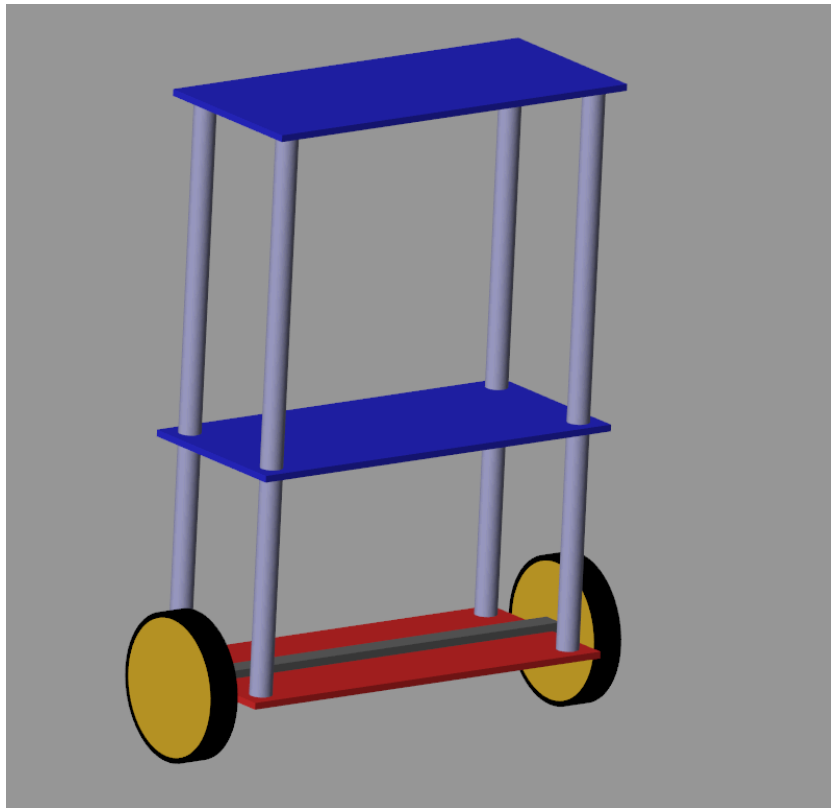


Fig: Robot Structure

Block Explanation and purpose

1)Solver Configuration: Solves mathematical equations related to the system.

2)World Frame: Create surroundings of the robot.

3)Mechanism Configuration: Defines physical parameters like gravity and relation between mechanical components.

4)Prismatic Joint: Models a connection that allows for linear motion, or translation, along a single axis. It constrains the two connected bodies to move relative to each other in a straight line.

5) Revolt Joint: Models a connection that allows for rotational motion about a single axis. It constrains the two connected bodies to rotate relative to each other around a common axis.

6)Rigid Transform Block: block defines a fixed spatial relationship between two coordinate systems, known as frames. Its primary purpose is to specify a constant translation and rotation from a base frame to a follower frame.

7)Different solid blocks are used to create the outer and inner wheel and the geometry of the robot.

8)External Force and Torque: Creates external force and torque in the simulation.

Simulation Procedures:

Creating the physical structure of the robot in simulation: Initially two wheels of the robot were created. The outer and inner part of the wheel were given different mass and color. To create this cylindrical block is used.

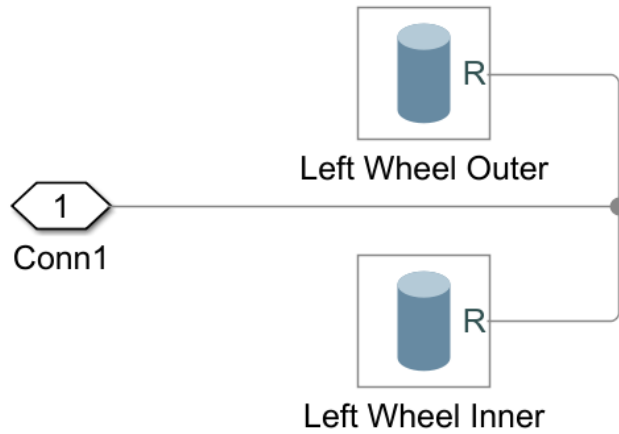


Fig: One Wheel

Then using another cylindrical block two wheels are mechanically connected. Rigid transform block is used here.

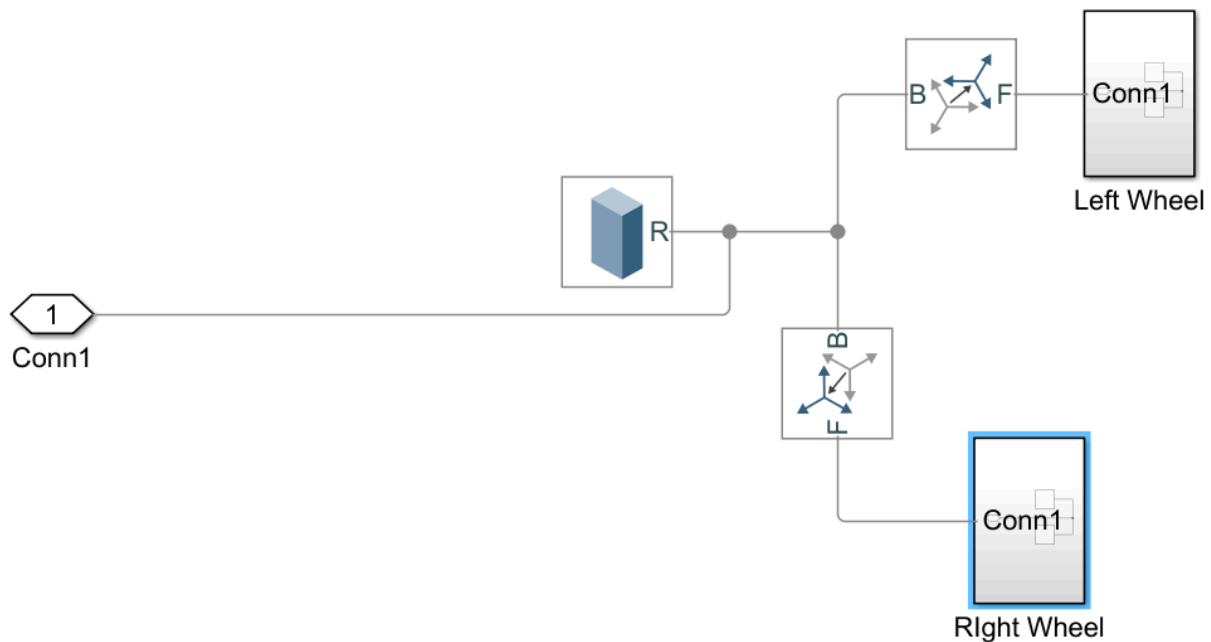


Fig: Two Wheels connected mechanically.

To create the chassis, 3 horizontal plates are made from rectangular blocks with proper dimensions. These plates are connected by 4 pillars. Using rigid transform blocks all plates and pillars are interconnected so that its motion becomes dependent on each other.

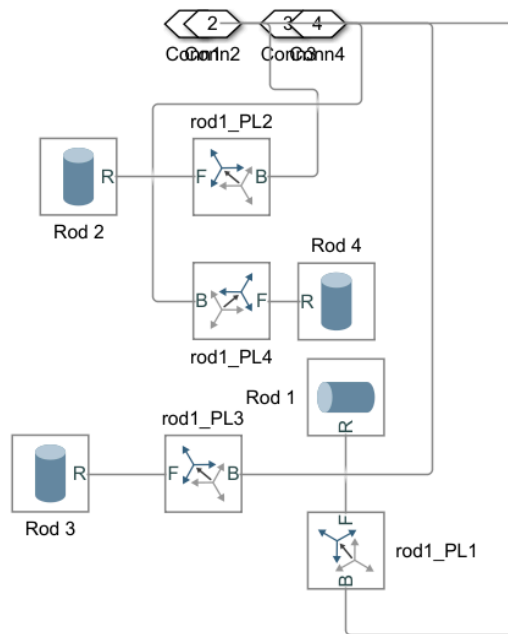


Fig: Interconnection of the pillars

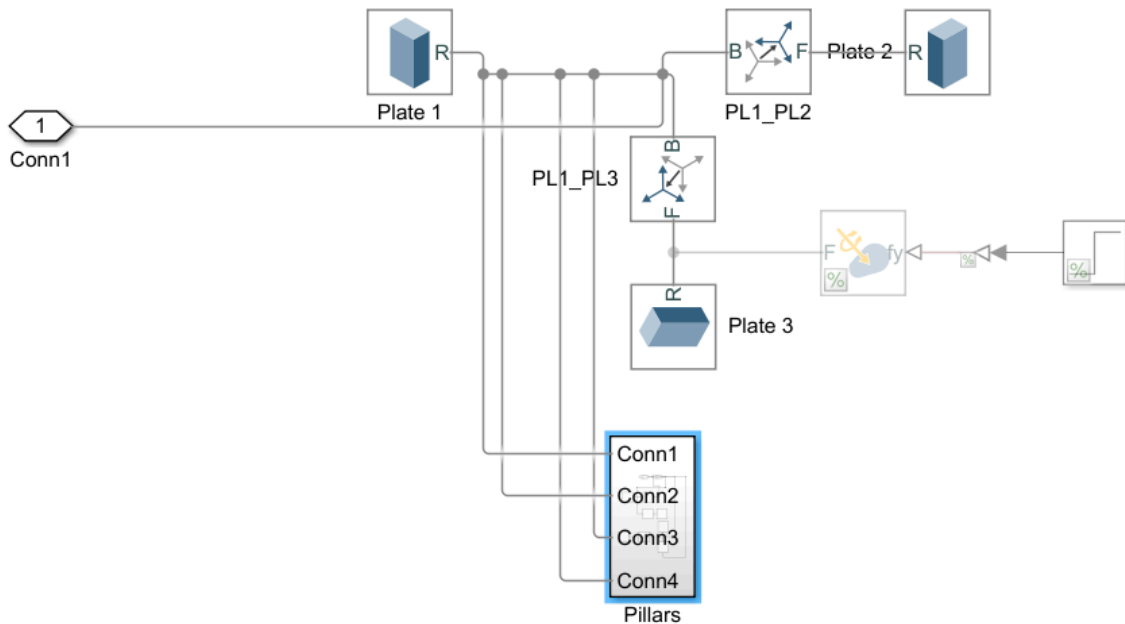


Fig: Pillars and the plates connection.

Chassis and cart(wheels) are connected by another rigid transform block. Then a PID controller is added. A step signal is given as an external force.

PID controller is connected to a prismatic joint balancing the robot. PID controller is tuned using the MATLAB app for tuning.

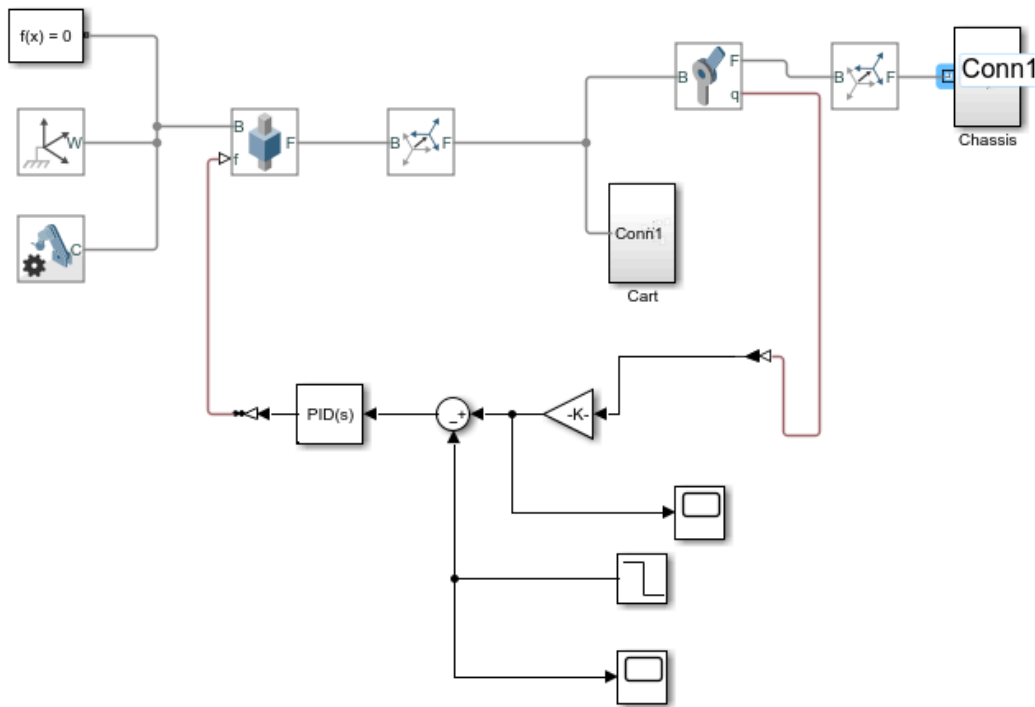


Fig: Overall Simulink Model

Working Principle:

When the step signal is applied on the robot, it tilts from the vertical line at a certain angle. Using the revolt joint this information is fed to the PID controller through a gain block and summing block. The summing block subtracts external force or step signal for this case from the tilted angle. Output of the summing junction is the error signal. PID controllers using different active devices reduce the error signal to stabilize the robot. This is fed to the joint which physically moves the robot to balance it.

We tuned PID controller to get the following response,

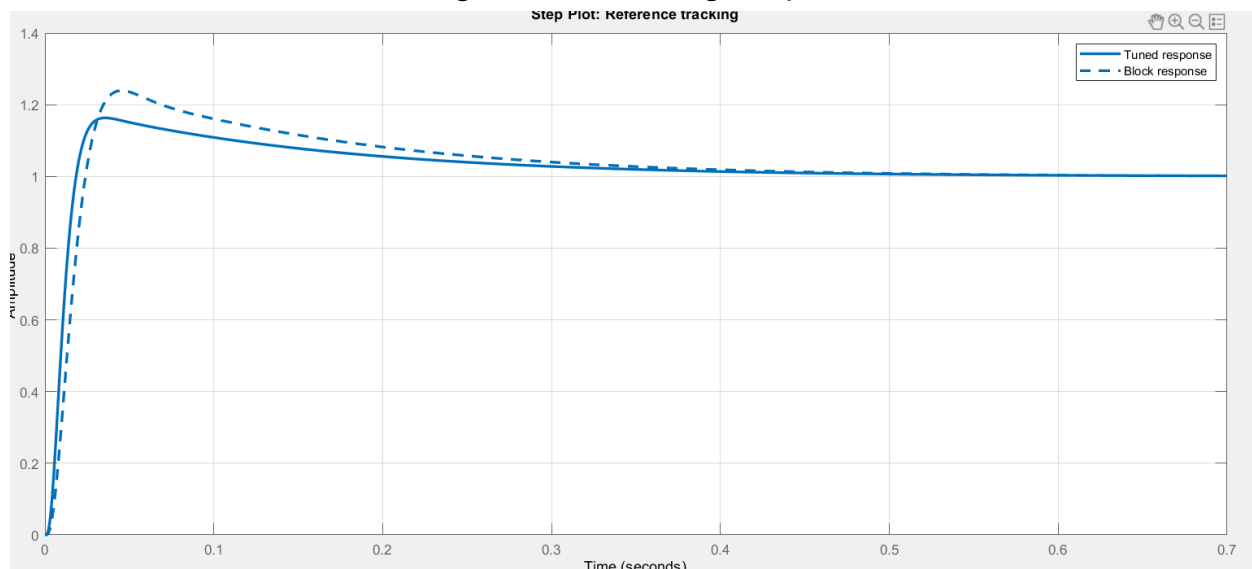


Fig: Response of PID

From the response we can see that there is very small overshoot, which physically represents the maximum tilt the robot will have. Also, it can be seen that there is no oscillation and the response goes to steady state pretty fast. This refers to the physical jerking being negligible apart from the initial shift. We got the best response using this value of the PID;

Proportional (P): -2.54414204727402

Integral (I): -9.30150763983751

Derivative (D): -0.165914829935758

Conclusion:

Using the abovementioned procedures, we successfully built a self balancing robot using a PID controller. The simulink simulation only establishes the foundation for the physical project, clearing our concepts of control system and different compensation techniques. Also this project required extensive knowledge of matlab and simulink.